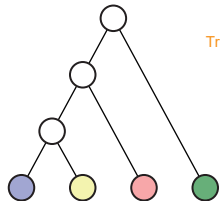


a Ideal triplet objective (ITO)

True rooted cell lineage tree



True rooted triplet relations

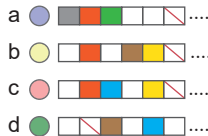
$a, b, c \rightarrow (a, b), c$
 $a, b, d \rightarrow (a, b), d$
 $a, c, d \rightarrow (a, c), d$
 $b, c, d \rightarrow (b, c), d$



All rooted triplets are correct and complete

b Partial triplet objective (PTO)

Observed terminal Barcodes(noisy)



Recorder noise sources



Finite recording capacity
(limited sites)



Dropout
(missing observation)



Homoplasy
(repeated mutation)

Reconstructed triplet statements

$a, b, c \rightarrow (b, c), a$ Missing/abstained

$a, b, d \rightarrow (a, d), b$ Incorrect

$a, c, d \rightarrow (a, c), d$ Correct

$b, c, d \rightarrow (b, c), d$ Correct

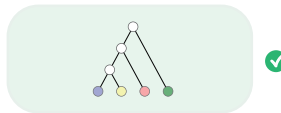
Correct triplet

Incorrect triplet

Missing/abstained

c Optimal triplet objective (OTO)

Candidate lineage trees



$$T_{\text{OTO}} \in \arg \max_T |S_{\text{tri}}(T) \cap S_{\text{tri}}(D)|$$

Maximum-agreement tree under the
observed partial triplet constraints